

APPENDIX FOR:
THE VALUE OF CLEAN WATER: EXPERIMENTAL EVIDENCE
FROM RURAL INDIA

Fiona Burlig, Amir Jina, and Anant Sudarshan

Contents

A	Additional context	A2
A.1	Taste testing	A2
A.2	Lab water quality testing	A3
B	Additional experiment details	A4
C	Experimental integrity	A7
C.1	Balance	A7
C.2	Compliance	A9
C.3	Exchangeable entitlement payments	A9
D	Additional results	A11
D.1	Sub-treatment-specific effects	A11
D.2	Survey evidence on clean water use	A12
D.3	Intent-to-treat effects on self-reported health	A13
D.3.1	Heterogeneous treatment effects	A14
D.4	Placebo effects on survey outcomes	A17
E	Pure control households vs. control households in treatment villages	A19
F	Calculating profitability	A22
G	Calculating cost-effectiveness	A24
H	Deviations from our pre-analysis plan	A27

A Additional context

A.1 Taste testing

We conducted a set of taste tests with five different water samples where the testers included 9 local survey enumerators. The purpose of these tests was to evaluate whether there was any *indicative* evidence in our setting for the commonly cited concern about the taste of water purified with point-of-use chlorine tablets or solution. These tests had separate IRB approval from the University of Chicago under Protocol No. IRB23-1363.

Subjects were asked to drink a cup of water labeled “Sample A,” (this was Bisleri, a leading bottled water brand) and think of it as a score of a 5 on a 1-10 taste scale. Then, subjects received four more blind water samples (“B” through “E”), presented in a randomized order, and asked to rate each sample on the same 1-10 scale. Respondents could return to Sample A whenever they wanted. The other samples were: Bisleri water where we added locally-available chlorine treatment solution added as per the dosing instructions on the packet; tap water with chlorine added; Spring Health water with chlorine added; and regular Spring Health water.

Appendix Figure A.1 plots the results of the taste test. Households rated all chlorinated sources at least 1 full point worse than Bisleri, and rated the treated water of our treatment (Spring Health) slightly better than Bisleri. 100% of respondents reported that one of the chlorinated water samples was their least preferred, and none reported that a chlorine-treated source was their most preferred.

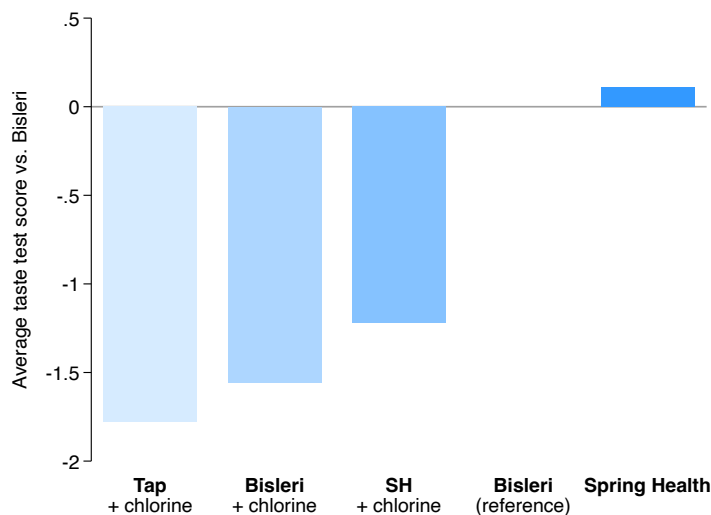


Figure A.1: Water taste tests

Notes: This figure plots the results from 9 water taste tests we conducted in Odisha (with separate IRB approval from the University of Chicago, IRB23-1363). All respondents were provided a sample of Bisleri, a standard bottled water. They were then provided four other sources in a randomized order: tap water treated with chlorine (per packet treatment instructions); Bisleri water treated with chlorine; Spring Health water treated with chlorine; and Spring Health without chlorine. Subjects were asked to compare each water source to Bisleri on a 1-10 scale, with Bisleri set to 5. Here, we plot the mean difference between the score of each source and the reference Bisleri. 100% of respondents ranked a chlorine-treated water source last.

Source: Burlig, Jina and Sudarshan (2025)

A.2 Lab water quality testing

We sent 19 water samples, drawn from our treatment villages, for water quality testing at the State Water Testing Laboratory R.W.S&S. This laboratory is used by the Odisha government for official water quality measurements. We collected 8 open well samples, 6 tube well samples, 4 tap water samples, and 1 Spring Health sample. Each sample was evaluated on heavy metals and biological contaminants. Appendix Figure A.2 plots the share of samples deemed “unacceptable” (i.e., worse than the “acceptable” threshold) across open wells, tube wells, taps, and Spring Health. None show evidence of heavy metal contamination. All but the Spring Health water show at least some unacceptable pH levels and biological contamination. Our implementation partner Spring Health also tests their water at regular intervals so this exercise was primarily intended to be informative about other sources.

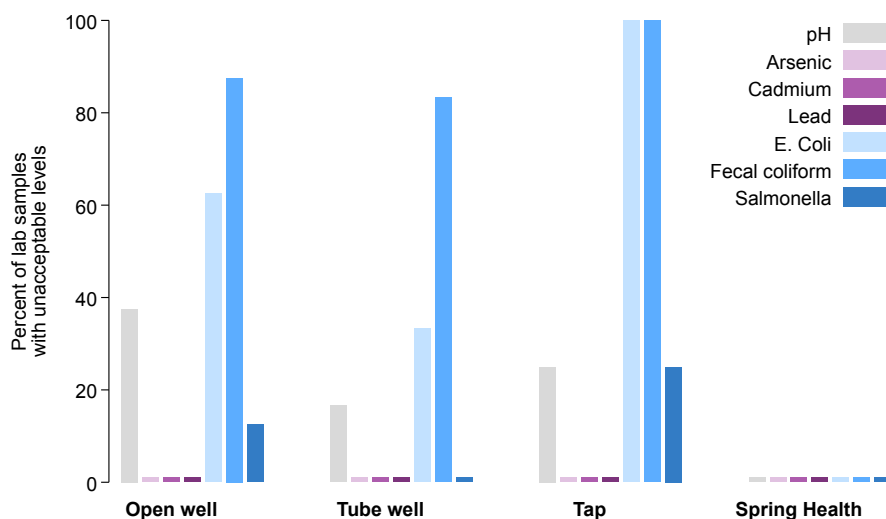


Figure A.2: Water lab testing results

Notes: This figure plots the results from testing water sampled in our experimental villages. Lab testing was conducted by the State Water Testing Laboratory R.W.S&S, Odisha, in Bhubhaneshwar. We collected 8 open well samples, 6 tube well samples, 4 tap water samples, and 1 Spring Health sample. Samples were classified as “unacceptable” per lab thresholds for each test. Bars show the share of samples deemed unacceptable for pH (gray), heavy metals (shades of purple), and biological contaminants (shades of blue).

Source: Burlig, Jina and Sudarshan (2025)

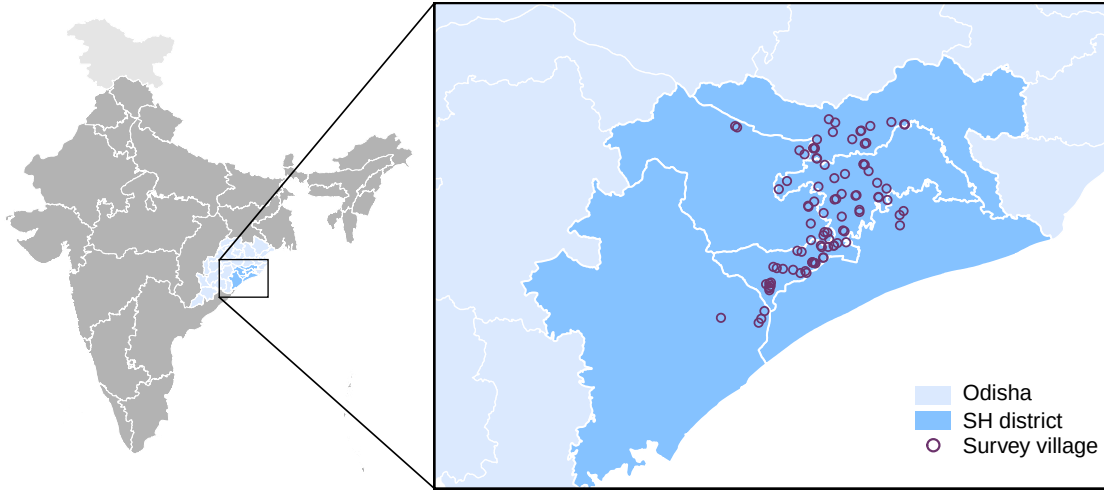


Figure B.1: Study village locations

Notes: This figure plots the location of our study villages within India. The light-blue-shaded districts plot the state of Odisha. Our experiment took place in the six dark-blue Spring Health districts plotted in the extruded view. The precise location of our experimental treatment villages (i.e., those where we conducted surveys, excluding the pure control) are denoted by purple circles.

Source: Development Data Lab (2025)

B Additional experiment details

This Appendix presents additional details about the experimental design and implementation. Appendix Figure B.1 plots the location of our study sample within India. The extruded view shows these districts in the context of Odisha, with the 120 treatment villages shown in orange circles.

Within each treatment village, we randomized households into sub-treatments or the control using scratch cards. The top panels of Appendix Figure B.2 shows a sample scratch card. All scratch cards were identical on the outside, save a unique ID number that linked scratch cards to treatment status. This mapping was known to the research team but not to the field staff. The bottom panel of Appendix Figure B.2 shows a photo of the scratch card in use.

Our experiment took place between May 2022 and October 2023. Appendix Figure B.3 shows the experiment timeline in the form of a Gantt chart. For logistical reasons, the sample was divided into phases or “implementation waves” (W1–W8 on the chart). At the start of each wave, scratch cards were delivered to households, with treatment offers beginning the following month. All treatment households were told their offers were valid for five months. As we obtained additional funding, at the end of these five months, we informed waves W3–W8 that their offers would be extended by two months. We also conduct five surveys (four short “high-frequency” checks designed to capture health and a longer endline) with a randomly-selected subset of households in each village. These are described in more detail in Section 4.

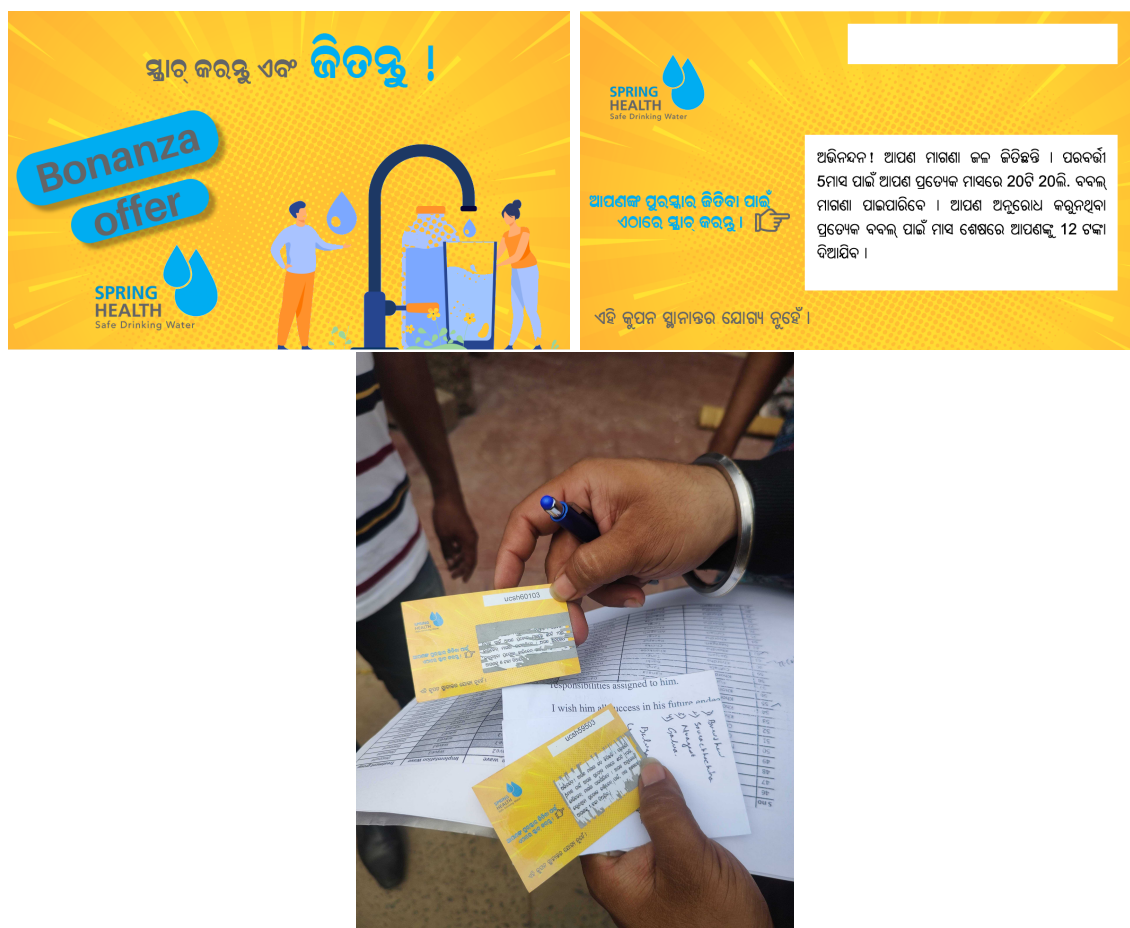


Figure B.2: Sample scratch card

Notes: This figure shows the promotional scratch card used to randomize households into treatments. The top panels show an example scratch card. On the left, we show the front of the card, which is common among all offers. On the right, we show the back of the card, which differs across offers (the white portion was hidden behind a scratch-off cover). The bottom panel shows a photograph of the real scratch cards. Each scratch card contained a unique ID, known to the research team but not to the field staff.



Figure B.3: Experimental timeline

Notes: This figure shows the experimental timeline in the form of a Gantt chart. W1–W8 refer to “implementation waves,” the staggered treatment roll-out. Randomization was stratified by wave. This figure shows when the scratch card distribution (and thus, the listing data collection) took place; the 5 (W1, W2) or 7 (all other waves) months of treatment offer validity; and the timing of all surveys. We conducted four high-frequency surveys (HFS-1 – HFS-4), and a longer endline survey for all villages in each wave.

C Experimental integrity

This section discusses the integrity of the experiment, including balance and compliance.

C.1 Balance

Due to logistical constraints with our partner organization, we were unable to conduct a baseline survey prior to the start of the experiment with our earliest survey conducted during the first month of offers. We therefore test for balance using data from our endline survey on demographic and other variables that we do not expect to change as a result of our treatments: household size, whether the household contains young children, the household head's education level, years the household head has lived in the village, and ownership of expensive appliances. Appendix Figure C.1 presents the means of these variables for each experimental group, p -values from pair-wise balance tests between each group. We find neither economically meaningful nor statistically significant differences between treatment arms on any variables. Because we check for balance using the endline data, the results presented in Appendix Figure C.1 account for attrition by construction.

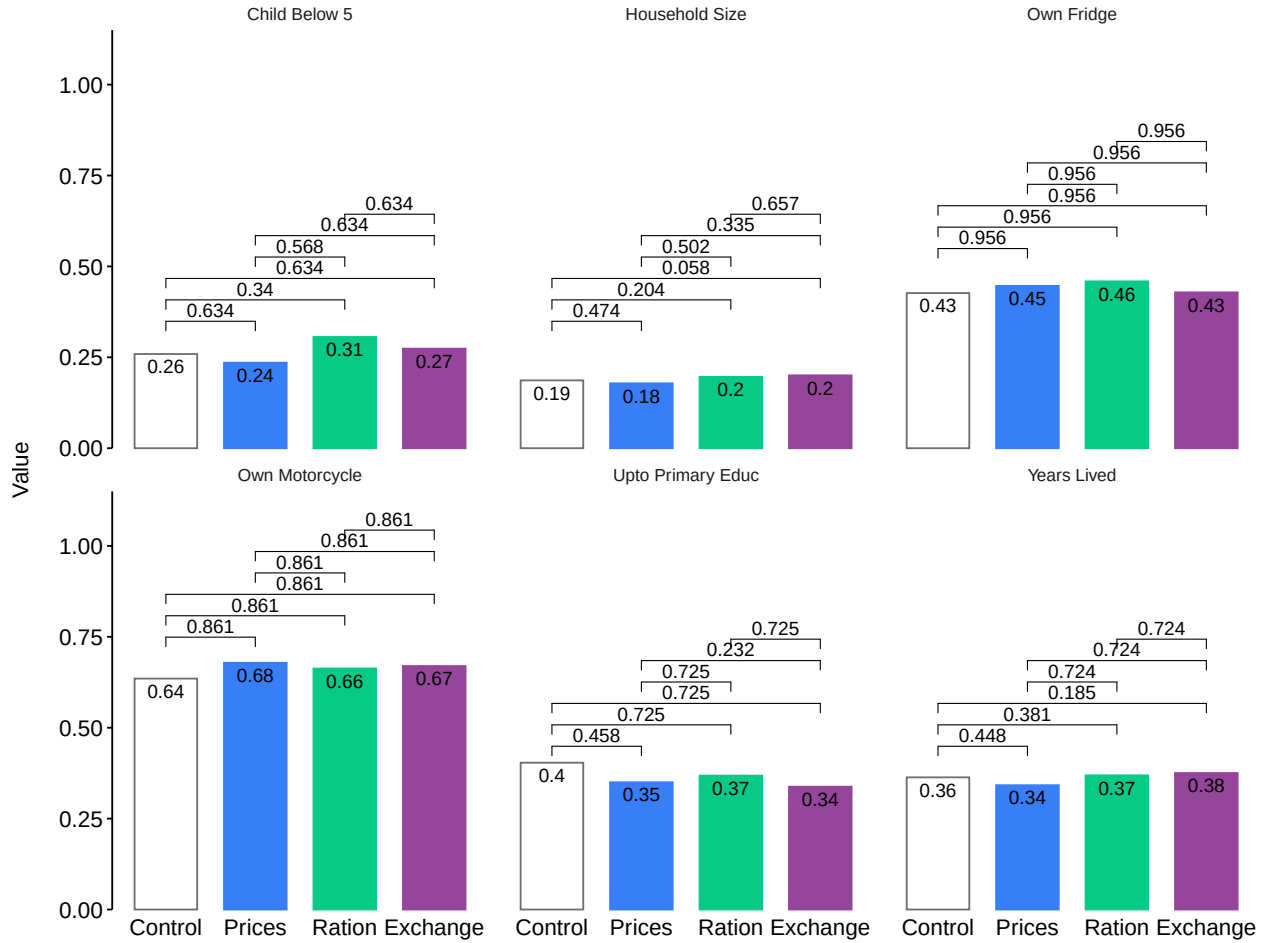


Figure C.1: Experimental balance

Notes: This figure shows experimental balance. Bars show average values for survey households in each treatment arm for six covariates. From top-left, these are: A dummy for the presence of a child below the age of 5 in the household, number of household members (with minimum and maximum values rescaled to lie between 0 and 1 for graphical convenience), a dummy for whether the household owns a refrigerator, a dummy for whether the household owns a two-wheeler/motorcycle, a dummy for whether the household head has at most a primary education, and the number of years the respondent has lived in the village (also rescaled for graphical convenience). We also conduct pairwise t -tests between each of the three main treatment arms and the control. Brackets show p -values with a Hommel adjustment for multiple comparisons.

C.2 Compliance

Midway through the experiment, our implementation partner had to deal with a period of increased operational and staffing challenges, partly caused by losses in revenue flow from sales of carbon credits, or by factors unrelated to the experiment such as flooding in some villages, wells running dry, or the franchise entrepreneurs operating the treatment arm choosing to quit. As a result in some months (and for some villages) the company temporarily paused operations. At these times water orders by all households are zero by definition. Figure C.2 shows that for the majority of villages, water delivery was available for 100% of scratch card offer months; across the sample, water delivery was available for 87.7% of village-months.

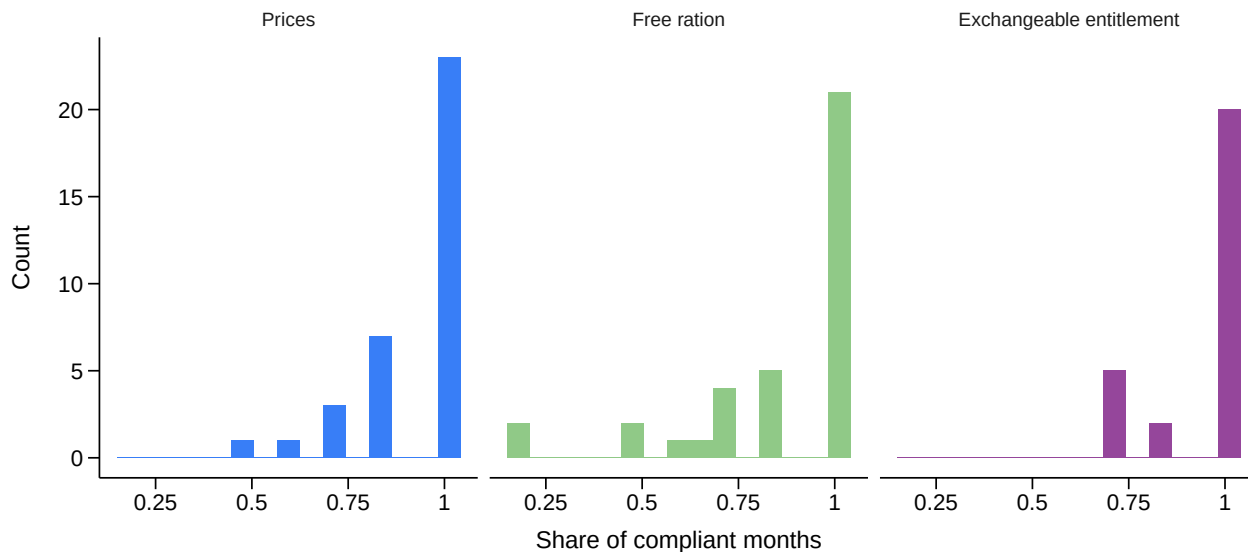


Figure C.2: Water delivery non-compliance

Notes: This figure plots histograms of the number of months households in each village were actually able to order water from Spring Health, divided by the intended offer duration, for each of our treatment arms. For example, households in a village with an offer duration of 7 months where Spring Health only operated for 5 months have a share of $\frac{5}{7} = 0.71$. Each observation is one village.

In our demand analysis, we exclude months where water was not delivered, as we are interested in households' take-up and usage of clean water under different allocation regimes, and months without a functioning seller reveal no information about demand for water or valuation. When conducting our cost-effectiveness analysis, we account for the fact that Spring Health deliveries did not occur 100% of the time.

C.3 Exchangeable entitlement payments

We might be concerned that households in the exchangeable entitlement arm chose to order water rather than receiving cash payments because they did not think they would actually be paid. This is a particular issue because while water arrives at the time of ordering, the rebate for which households were eligible could only be calculated at the end of the month.

In addition, because transfers were often very small, our implementation partner sometimes clubbed payments for multiple months together creating delays.

If trust is a concern here then we would expect that households would change their ordering behavior following the first payment since this transfer makes it clear that the rebate offer was real. We therefore carry out an event study style analysis, using only exchangeable entitlement households to estimate the equation below:

$$Y_{it} = \sum_{d=-30}^{30} \beta_d 1[\text{Days to payment} = d]_{it} + \alpha_i + \varepsilon_{it} \quad (\text{C.1})$$

where Y_{it} are water orders by exchangeable entitlement household i on date t . $1[\text{Days to payment} = d]_{it}$ is an indicator equal to 1 if household i is d days from payment on date t , α_i are household fixed effects, and ε_{it} is an error term, clustered at the village level. We restrict the sample to days around the first time households are paid, to avoid contamination of the pre-period in subsequent months, and because the first payment is where we are most likely to see trust related effects. Appendix Figure C.3 plots the results. We find no evidence of changes in water ordering behaviour at the timing of payment.

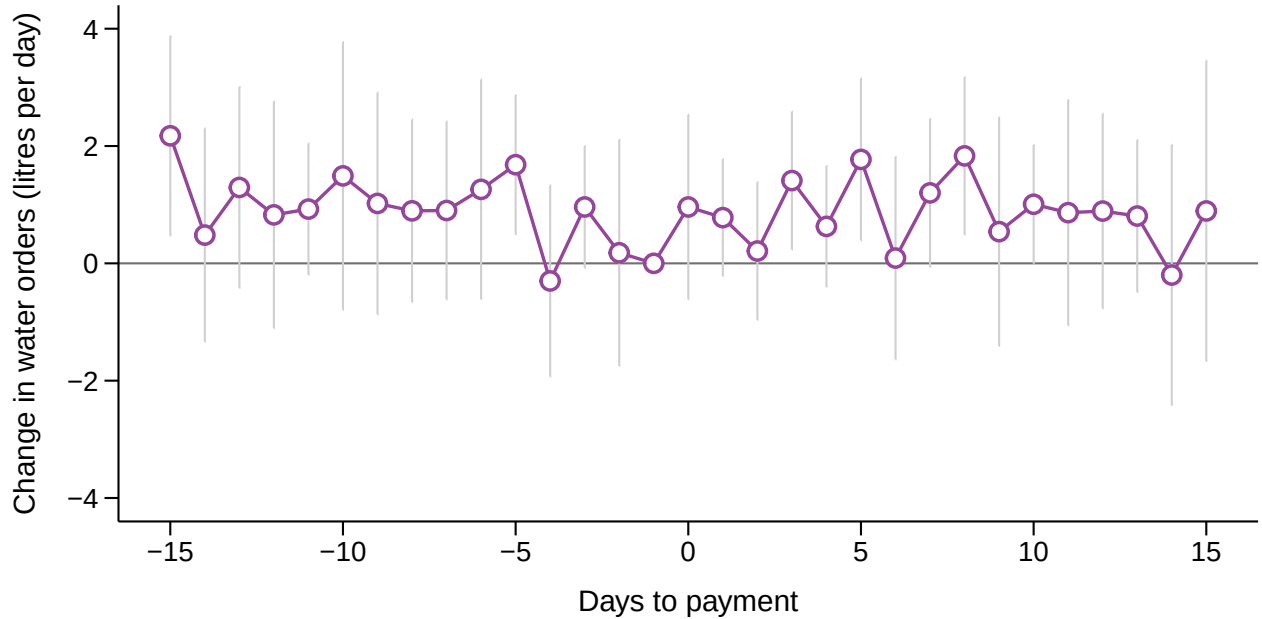


Figure C.3: Effect of payment on water orders in the exchangeable entitlement arm

Notes: This figure plots changes in daily water orders among exchangeable entitlement households only, relative to 1 day before the timing of the first rebate payment. We estimate these coefficients using Equation C.1, which includes household fixed effects. Confidence intervals are derived from standard errors which are clustered at the village level.

D Additional results

D.1 Sub-treatment-specific effects

Dynamic effects by sub-treatment Figure 4 plots the effects of treatment on water prices as a function of months since the beginning of treatment. In this main text figure, we pool across sub-treatments. Appendix Figure D.1 presents results for each sub-treatment separately. As in Figure 4, we find that demand is very stable across time for all sub-treatments.

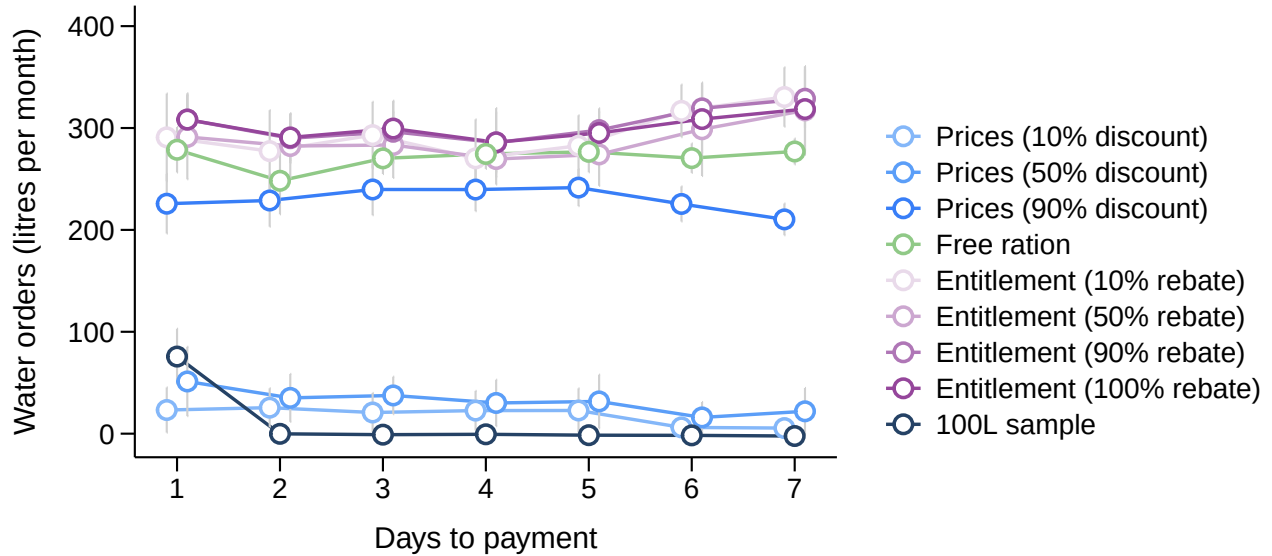


Figure D.1: Water orders event study (unpooled)

Notes: This figure plots the effect of our treatments on water orders as a function of time since treatment started (with the first month offers were active set to 1), estimated using a sub-treatment-specific version of Equation (2). Standard errors are clustered at the village level. 35 villages — the first enrolled in the experiment — had only 5 months of treatment, while the remainder had 7. The sample is restricted to months when deliveries occurred.

Self-reported health effects by sub-treatment Appendix Table D.1 the intent-to-treat effects of offers on self-reported health outcomes using a version of Equation (3) that estimates a separate coefficient for each sub-treatment arm. Point effects are negative across self-reported sickness, health expenses, and missed work outcomes. Standard errors are larger with the most precisely estimated effects on self-reported sickness and missed work measures in the discount group although we cannot reject equality of treatment effects across most treatment arms. Nevertheless, it is interesting that the sub-group facing the highest price signal — namely the 10% discount group — also shows the largest treatment effects on these outcomes, suggestive of high prices perhaps inducing screening behaviour or impacts on complementary sanitation activities.

Table D.1: Effect of treatment offers on self-reported health, unpooled

	Sickness	Health Expenses	Missed Work	Symptoms				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
10% Discount	-0.14 (0.05)	-32.58 (28.51)	-0.08 (0.02)	-0.03 (0.01)	-0.02 (0.02)	-0.03 (0.02)	-0.02 (0.01)	-0.06 (0.02)
50% Discount	-0.11 (0.05)	-20.67 (31.99)	-0.03 (0.03)	-0.03 (0.01)	-0.03 (0.03)	-0.02 (0.02)	-0.03 (0.01)	-0.03 (0.03)
90% Discount	-0.09 (0.06)	-2.72 (32.35)	-0.02 (0.03)	-0.02 (0.01)	-0.02 (0.02)	-0.02 (0.02)	0.00 (0.01)	-0.03 (0.03)
10% Rebate	-0.11 (0.10)	8.80 (56.41)	-0.08 (0.07)	-0.02 (0.03)	-0.06 (0.04)	-0.04 (0.02)	-0.02 (0.04)	-0.01 (0.04)
50% Rebate	-0.13 (0.11)	-87.78 (79.75)	-0.09 (0.06)	-0.04 (0.02)	-0.07 (0.04)	-0.04 (0.04)	-0.03 (0.03)	-0.03 (0.04)
90% Rebate	-0.19 (0.08)	-86.53 (58.70)	-0.15 (0.05)	-0.03 (0.02)	-0.07 (0.04)	-0.06 (0.03)	-0.02 (0.03)	0.00 (0.04)
100% Rebate	-0.12 (0.09)	-75.82 (56.28)	-0.06 (0.05)	-0.03 (0.02)	-0.04 (0.03)	-0.03 (0.03)	-0.02 (0.03)	-0.04 (0.04)
Free Ration	-0.07 (0.06)	-30.46 (37.03)	-0.05 (0.03)	0.00 (0.01)	-0.01 (0.02)	-0.03 (0.02)	0.00 (0.01)	-0.02 (0.03)
N	4,670	4,433	4,670	4,670	4,670	4,670	4,670	4,670
Control Means	0.516	239.507	0.2	0.044	0.159	0.108	0.053	0.223

Notes: This table presents intent-to-treat effects of our treatment offers on health outcomes, estimated using Equation (3), and unpooled to provide estimates for each sub-offer arm of the main treatment arm. We restrict the sample to months when water delivery occurred and drop the top 1 percent of health expenses to remove large outliers. In column (1), the outcome is an indicator for the number of household members being sick in the past week. In column (2), the outcome is household spending on health in the past week in INR. In column (3), the outcome is an indicator for the number of household members missing work due to illness in the past week. In columns (4)–(8), the outcome is an indicator for a household member reporting symptoms of vomiting, fever, stomach ailments (gastric pain/abdominal pain/diarrhea), flu symptoms (cough/congestion/headache/fatigue), and other symptoms (e.g., skin infection, joint pains, etc). All regressions include village and month-of-sample fixed effects. Standard errors are clustered by village.

D.2 Survey evidence on clean water use

We use our survey data to corroborate our demand results. Using Equation (3), and a version of this equation where we break treatment effects out by offer level for the price and exchange arms, we estimate the effect of our water treatment offers on the number of drinking water sources used by the household, whether the household drinks any clean water, and whether the household drinks only clean water. Appendix Table D.2 reports the results. We find large and significant impacts of all treatments on all three outcome variables. Treatment effects grow with discount size in the price arm, but do not meaningfully vary with rebate size in the exchangeable entitlement arm. Note that Column (3) also serves as a first stage for the IV regressions we present in Section 5.4. We restrict data to months where water was sold and to households reporting at least one drinking water source since some households did not report any sources during one or more survey rounds.

Table D.2: Effects of treatment on water use: survey data

	Number of sources		Drinks clean water		Only drinks clean water	
	(1)	(2)	(3)	(4)	(5)	(6)
Discount	0.34 (0.06)		0.34 (0.03)		0.09 (0.02)	
Exchange	0.75 (0.09)		0.77 (0.04)		0.25 (0.03)	
Free Ration	0.61 (0.09)	0.61 (0.09)	0.63 (0.04)	0.63 (0.04)	0.17 (0.03)	0.17 (0.03)
10% Discount		0.26 (0.06)		0.30 (0.04)		0.09 (0.03)
50% Discount		0.23 (0.09)		0.24 (0.04)		0.07 (0.02)
90% Discount		0.51 (0.12)		0.48 (0.05)		0.10 (0.03)
10% Rebate		0.71 (0.13)		0.76 (0.04)		0.23 (0.04)
50% Rebate		0.72 (0.11)		0.78 (0.04)		0.24 (0.04)
90% Rebate		0.66 (0.10)		0.75 (0.05)		0.29 (0.05)
100% Rebate		0.85 (0.12)		0.78 (0.05)		0.24 (0.03)
N	5,285	5,285	5,276	5,276	5,285	5,285
Control Means	1.443	1.443	0.004	0.004	0.001	0.001

Notes: This table presents intent-to-treat effects of water offers on consumption of Spring Health water, estimated using Equation (3) in Columns (1), (3), and (5), and using a version of this same equation with effects broken out by sub-treatment arm in Columns (2), (4), and (6). Columns (1) and (2) present the effect on the number of drinking water sources, Columns (3) and (4) report drinking any Spring Health water, and Columns (5) and (6) report only drinking Spring Health water. We restrict the sample to village-months where Spring Health delivered water. All regressions include village and month-of-sample fixed effects. Standard errors are clustered at the village level.

D.3 Intent-to-treat effects on self-reported health

Appendix Table D.3 reports intent-to-treat effects on self-reported health outcomes. The first three columns show impacts on overall health: Column 1 presents treatment effects on the number of household members reported being sick in the past week; Column 2 presents treatment effects on weekly health expenditures; and Column 3 reports treatment effects on whether any household member had to miss work due to sickness in the past week. Average illness in this population is high, with 52% of individuals in control households reporting having been sick in the last week; mean health spending in the control group of INR 240 (10% of average expenditure), and 20% of control households reporting that a household member needed to miss work due to illness in the past week.

We see meaningful improvements in these self-reported measures of health, though our estimates are not always precise. The point estimates of measures of reported sickness correspond to reductions of 21% (FDR-adjusted $p = 0.02$), 27% (FDR-adjusted $p = 0.15$),

and 14% (FDR-adjusted $p = 0.29$) in the price arm, exchangeable entitlement arm, and free ration arm, relative to control. The estimated effects on weekly health expenses are all very noisy, though point estimates are negative and imply meaningful reductions in monthly spending – between 17% and 61% of our estimated WTA.²⁷ Finally, we estimate substantial declines in missed work, though again, these are only different from zero at conventional levels in the prices arm: 20% (FDR-adjusted $p = 0.05$), 45% (FDR-adjusted $p = 0.15$), and 25% (FDR-adjusted $p = 0.24$) for prices, entitlements, and rations, respectively.²⁸ Finally, columns 4 through 8 present results on various individual symptoms. Because our goal was to measure treatment effects on water-borne disease, all symptoms included in the survey could have plausibly been improved by clean water; we did not include placebo measures such as broken limbs in the survey. Though these treatment effects are imprecisely measured, the broad pattern of point estimates suggests weak evidence of reductions in specific illnesses.

Taken together, our intent-to-treat results corroborate our household demand estimates: households report experiencing meaningful improvements in health as a result of our clean water offers. Because our offers broadly move households on the extensive, rather than intensive, margin of clean water use, to understand the effect of clean water access on health requires estimating a local average treatment effect. Put differently, simply scaling the results in Appendix Table D.3 by quantities ordered in litres (Column (3) of Table 1) does not yield a dose-response function, because differences in quantities across treatment arms are driven by selection into consumption. This is directly addressed in the instrumental variables estimates in the main text.

In Appendix D.4, we take advantage of the occasional service disruptions discussed in Section 3.6 to carry out a further robustness check, re-estimating these specifications using only observations from months when water sales were disrupted. If our health effects are indeed driven by access to clean drinking water, they should be significantly attenuated or disappear entirely in months where households could not obtain this water. Appendix Table D.5 demonstrates that this is broadly the case. We also do not find evidence that self-reported health meaningfully *worsens* during these service disruptions, suggesting that disruptions are not more costly than the absence of clean water itself, though with the caveat that this test relies on a relatively small and selected sub-sample of villages.

D.3.1 Heterogeneous treatment effects

We test for heterogeneous treatment effects on self-reported health (across all treatment arms) using the following specification:

$$\begin{aligned}
Y_{it} = & \beta_1 \text{Any discount}_i + \beta_2 \text{Any discount}_i \times \text{Covariate}_i \\
& + \beta_3 \text{Any exchange}_i + \beta_4 \text{Any exchange}_i \times \text{Covariate}_i \\
& + \beta_5 \text{Free ration}_i + \beta_6 \text{Free ration}_i \times \text{Covariate}_i \\
& + \text{Covariate}_i + \gamma_v + \delta_t + \varepsilon_{it}
\end{aligned} \tag{D.1}$$

²⁷Regression estimates are, per our survey question, on health spending in the past week. We thus scale by 4 when comparing to our estimated WTA for monthly clean water access.

²⁸We did not test for infant mortality, as our sample is too small to detect effects on this rare outcome (Kremer et al., 2023).

Table D.3: Intent-to-treat effect of water offers on self-reported health

	Sickness	Health expenses	Missed work	Symptoms				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Prices (Discounts)	-0.11 (0.04) [0.02]	-18.25 (21.90) [0.41]	-0.04 (0.02) [0.05]	-0.03 (0.01)	-0.02 (0.02)	-0.02 (0.01)	-0.02 (0.01)	-0.04 (0.02)
Exchangeable entitlements	-0.14 (0.08) [0.15]	-64.49 (53.87) [0.23]	-0.09 (0.05) [0.15]	-0.03 (0.02)	-0.06 (0.03)	-0.04 (0.03)	-0.02 (0.03)	-0.02 (0.03)
Free ration	-0.07 (0.06) [0.29]	-30.39 (37.00) [0.41]	-0.05 (0.03) [0.24]	0.00 (0.01)	-0.01 (0.02)	-0.03 (0.02)	0.00 (0.01)	-0.02 (0.03)
N	4,670	4,433	4,670	4,670	4,670	4,670	4,670	4,670
Control Means	0.516	239.507	0.2	0.044	0.159	0.108	0.053	0.223

Notes: This table presents intent-to-treat effects of our treatment offers on health outcomes, estimated using Equation (3). In column (1), the outcome is the number of household members being sick in the past week. In column (2), the outcome is household spending on health in the past week in INR. In column (3), the outcome is an indicator for the number of household members missing work due to illness in the past week. In columns (4)–(8), the outcome is an indicator for a household member reporting symptoms of vomiting, fever, stomach ailments (gastric pain/abdominal pain/diarrhea), flu symptoms (cough/congestion/headache/fatigue), and other symptoms (e.g., skin infection, joint pain, etc.). We restrict the sample to village-months where Spring Health delivered water, and drop the top 1 percent of health expenses to remove large outliers. All regressions include village and month-of-sample fixed effects. Standard errors are clustered by village. FDR-adjusted p -values in brackets.

where Y_{it} is an indicator for any ill household member in the previous week, spending on health in the past week, or an indicator for any household member having missed work in the past week due to illness. Covariate $_i$ are household characteristics: an indicator for children below 5 in the household; a dummy for whether the household head completed at most primary education; a dummy for ever reporting the use of piped water during any survey round; and a dummy for above-median income; above- vs. below-median monthly household income (binned) in the first survey. γ_v and δ_t are village and month-of-sample fixed effects, and ε_{it} is an error term, clustered at the village level. We restrict the sample to households in price treatment villages.

Appendix Table D.4 presents heterogeneity in health intent-to-treat effects. We see statistically significant evidence that the benefits of treatment in the Prices arm on the various self-reported health outcomes are concentrated amongst households with small children. Health expenses also slightly higher amongst higher-income households. We view the first of these facts as further evidence demonstrating that our health effects indeed result from clean water access, as children are likely the most susceptible to diseases such as diarrhoea, and their illnesses can plausibly lead to parents missing work.

Table D.4: Heterogeneous effect of offers on self-reported health outcomes

	Sickness (1)	Expense (2)	Missed work (3)	Sickness (4)	Expense (5)	Missed work (6)	Sickness (7)	Expense (8)	Missed work (9)	Sickness (10)	Expense (11)	Missed work (12)
Prices (Discounts)	0.17 (0.17)	306.03 (122.27)	0.25 (0.16)	-0.10 (0.05)	-20.12 (23.71)	-0.04 (0.02)	-0.12 (0.06)	-22.36 (26.25)	-0.03 (0.03)	-0.10 (0.06)	-4.17 (26.74)	-0.04 (0.03)
Exchangeable entitlements	0.30 (0.34)	365.66 (239.17)	0.26 (0.34)	-0.19 (0.10)	-109.04 (63.44)	-0.13 (0.06)	-0.16 (0.09)	-67.84 (65.99)	-0.10 (0.05)	-0.08 (0.08)	-30.63 (50.76)	-0.07 (0.05)
Free ration	0.09 (0.18)	-24.85 (107.84)	0.07 (0.08)	-0.08 (0.06)	-30.25 (38.07)	-0.06 (0.03)	-0.06 (0.07)	-21.11 (51.72)	-0.01 (0.03)	-0.06 (0.08)	-42.93 (39.46)	-0.04 (0.04)
Covariate	0.02 (0.13)	22.61 (85.19)	0.02 (0.07)	0.00 (0.07)	-23.15 (37.69)	-0.04 (0.03)	0.01 (0.05)	-23.78 (41.29)	0.03 (0.03)	0.09 (0.06)	37.46 (35.76)	0.01 (0.04)
Discount $\times$ Covariate	-0.31 (0.18)	-310.75 (127.63)	-0.28 (0.15)	-0.02 (0.09)	11.13 (47.37)	0.00 (0.04)	0.01 (0.08)	-22.18 (43.10)	-0.04 (0.04)	-0.05 (0.08)	-79.06 (42.67)	-0.03 (0.04)
Exchange $\times$ Covariate	-0.49 (0.35)	-355.59 (246.92)	-0.42 (0.34)	0.20 (0.11)	141.32 (63.91)	0.10 (0.05)	0.04 (0.08)	-26.60 (59.89)	0.01 (0.05)	-0.19 (0.08)	-116.88 (56.57)	-0.08 (0.04)
Ration $\times$ Covariate	-0.14 (0.20)	28.99 (103.31)	-0.11 (0.10)	0.04 (0.10)	-4.83 (64.33)	0.05 (0.05)	-0.01 (0.08)	-17.09 (63.25)	-0.09 (0.05)	-0.03 (0.10)	28.77 (51.08)	-0.02 (0.05)
N	1,405	1,364	1,405	4,681	4,445	4,681	4,512	4,282	4,512	4,512	4,282	4,512
Covariate		Child below 5			Low education			Piped water			High income	

Notes: This table presents intent-to-treat effects of water offers on water orders at the monthly level for the prices arm, with discount levels interacted with time-invariant covariates, estimated using Equation (??). The outcome variable in all columns is total monthly water order quantity in litres. The covariates we include are: (i) a dummy for the presence of a child below 5, (ii) a dummy for the household head having at most primary education, (iii) a dummy for ever reporting the use of piped water in any survey round, and (iv) a dummy for having above-median income. We restrict the sample to price group villages only, and include only village-months where Spring Health delivered water. All regressions include village fixed effects. Standard errors are clustered at the village level.

D.4 Placebo effects on survey outcomes

In Appendix C we present data on non-compliance during the experiment, showing that water was not delivered (sold) by our implementation partner in only 12.3% of village-months during our sample period. Our broad policy in the field was not to conduct surveys in these village-months, both because households could not place orders and in some cases because our partner advised us about household sensitivities where they had suddenly ceased operations. That said, we did conduct a small number of surveys in these months, typically because we were not informed of operational disturbances in advance.

We thus run a set of regressions using Equation (3), restricting the sample to the village-months where water distribution was halted. With the (significant) caveats that the disruption itself could affect household responses and the fact that these village-months are not representative of the broader sample, this exercise provides a useful placebo test of the impact of clean water on health.

Table D.5 presents the effects of water offers on self-reported health during months when Spring Health deliveries were unavailable, broken out by sub-treatment arm. We see no evidence of improvements in self-reported health outcomes (i.e., negative treatment effects) in these months. For the exchangeable entitlement and free ration groups, the point estimates appear attenuated, albeit noisy. Perhaps surprisingly, we see a *positive* effect on health outcomes (i.e., households report being *sicker* than the control) in the 90% discount group. However, we see no similar effects in the other discount groups. Thus, it is likely that this is driven by idiosyncratic differences in health outcomes in a few households given the relatively small sample size we are left with in this regression (393 household-month observations with 6 discount group villages, 4 exchangeable entitlement villages, and 5 free ration villages).

Table D.5: Placebo intent-to-treat effect of water offers on self-reported health

	Sickness	Health Expenses	Missed Work	Symptoms				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
10% Discount	0.04 (0.07)	-47.86 (47.25)	0.04 (0.03)	-0.01 (0.03)	-0.05 (0.03)	0.07 (0.06)	0.00 (0.00)	-0.02 (0.05)
50% Discount	0.03 (0.13)	119.90 (131.35)	0.06 (0.04)	-0.01 (0.02)	-0.02 (0.07)	0.14 (0.06)	0.03 (0.03)	-0.01 (0.02)
90% Discount	0.26 (0.09)	265.76 (108.50)	0.28 (0.09)	-0.03 (0.02)	0.11 (0.06)	0.12 (0.05)	0.03 (0.02)	-0.03 (0.06)
10% Rebate	0.03 (0.12)	19.63 (73.29)	-0.09 (0.06)	0.00 (0.00)	-0.02 (0.10)	0.01 (0.07)	0.01 (0.07)	-0.01 (0.09)
50% Rebate	-0.32 (0.18)	-106.05 (58.69)	-0.09 (0.06)	0.00 (0.00)	-0.18 (0.13)	-0.05 (0.05)	-0.05 (0.05)	-0.19 (0.10)
90% Rebate	0.18 (0.33)	158.72 (281.73)	0.44 (0.12)	0.00 (0.00)	0.02 (0.16)	0.04 (0.02)	-0.05 (0.05)	-0.03 (0.13)
100% Rebate	-0.09 (0.27)	-157.64 (102.84)	0.03 (0.10)	0.08 (0.05)	-0.14 (0.13)	0.03 (0.10)	-0.05 (0.05)	-0.15 (0.13)
Free Ration	-0.18 (0.19)	-74.84 (84.95)	-0.01 (0.10)	0.00 (0.02)	0.02 (0.07)	-0.08 (0.08)	0.05 (0.02)	-0.08 (0.07)
N	393	361	393	393	393	393	393	393
Control Means	0.423	250.417	0.109	0.044	0.131	0.102	0.007	0.197

Notes: This table presents intent-to-treat effects of our treatment offers on health outcomes, estimated using Equation (3), on only the sample of village-months where water distribution was halted. In column (1), the outcome is the number of household members being sick in the past week. In column (2), the outcome is household spending on health in the past week in INR. In column (3), the outcome is an indicator for the number of household members missing work due to illness in the past week. In columns (4)–(8), the outcome is an indicator for a household member reporting symptoms of vomiting, fever, stomach ailments (gastric pain/abdominal pain/diarrhea), flu symptoms (cough/congestion/headache/fatigue), and other symptoms (e.g., skin infection, joint pain, etc.). We drop the top 1 percent of health expenses to remove large outliers. All regressions include village and month-of-sample fixed effects. Standard errors are clustered by village.

E Pure control households vs. control households in treatment villages

In this Appendix, we compare household water orders in pure control villages to water orders in control households in treatment villages. We restrict the sample to households who place orders for water, because while we have a full listing of potential customers in the treatment villages (i.e., we observe households both ordering and not ordering water), in the pure control villages, we only observe households who order water.

To carry out this comparison, we first run a regression of the following type on pure control and control consumers:

$$Y_{it} = \alpha + \beta_1 \cdot 1[\text{Treatment village}]_v + \delta_m + \varepsilon_{vt} \quad (\text{E.1})$$

where Y_{it} is the total water orders or the mean order size of consumer i in month-of-sample t , $1[\text{Treatment village}]_v$ is a dummy that takes the value 1 if consumer i is a control household in one of the treatment villages and 0 if they are a consumer in a pure control village, δ_m is a month fixed effect, and ε_{vt} is an error term, clustered at the household level.

We also run a variant of this regression to compare the number of unique control-group consumers in treatment vs. pure control villages at the village-month level as below:

$$N_{vt} = \alpha + \beta_1 \cdot 1[\text{Treatment village}]_v + \delta_t + \varepsilon_{vt} \quad (\text{E.2})$$

where N_{vt} are the total number of consumers in village v in month t , counting only control consumers in treatment villages and all consumers in pure control villages. $1[\text{Treatment village}]_v$ is a dummy that takes the value 1 if village v is a treatment village and 0 if it is a pure control village, δ_m is a month fixed effect, and ε_{vt} is an error term, clustered at the household level.

Because both types of consumers face exactly the same price, we might expect no difference in water orders. In practice, however, this is not the case. Table E.1 presents the results. We find that consumers in the pure control villages (where no experiment activities or surveys took place) order more water (column 1) and are more numerous (column 3) than control condition consumers in treatment villages. However, we find that the average order size in the two groups is similar (column 2), suggesting that pure control households order water more often.²⁹

There are two explanations for this divergence. The first is that there are spillovers induced by the treatment. Such spillovers could occur if households facing full price in villages where others have been given special offers react by reducing the probability of placing an order and reducing how much they order. This type of spillover would lead us to *underestimate* the WTP for water, because it would dampen the demand observed at

²⁹Some of the gap in the consumer count would occur because there is a smaller pool of controls in experiment villages since 39 households were assigned to treatment offer. Because the size of villages is large this is a small effect. The average population of villages in our sample is 460 households, balanced across conditions by randomization. Based on this we could adjust the true consumer count difference to 22 consumers instead of 24.

Table E.1: Water orders, control households in pure control vs. treatment villages.

	Total orders (1)	Average order (2)	Consumers (3)
Experiment Village Control	-45.12 (7.39)	-0.13 (0.99)	-24.32 (1.52)
N	14,501	14,501	638
Reference (Pure Control) Means	293.51	12.36	36.97

Notes: This table presents tests for differences between all consumers in Pure Control villages vs. control-condition consumers in other (treatment) experiment villages following Equations E.1 (columns 1 and 2) and E.2 (column 3). We only include households that ever ordered positive quantities of water. The outcome in (1) is mean total monthly orders in litres, (2) is the average order size in litres, and (3) is the total number of non-zero consumers. Regressions include month fixed effects, and standard errors are clustered at the village level.

the highest price, relative to what might have been observed in the absence of treatment households. It would also lead us to slightly overestimate the intent-to-treat effects on health outcomes, though the instrumental variable estimates should not be affected.

The other explanation for the divergence is that our implementation partner did not market as intensively to control consumers in the treatment villages as they did to households in the pure control villages. A difference in sales *effort* would render the comparison of these two groups uninformative about behavioural spillovers, since full-price consumers would no longer face similar seller behaviour.

We do not have direct measures of “effort,” but there is suggestive evidence that this is the more likely explanation. The first piece of evidence comes from a comparison of control water orders across the three arms of the experiment. Specifically, we compare water orders across our three treatment village types among control households only:

$$Y_{it} = \pi_1 1[\text{Exchangeable entitlement village}]_v + \pi_2 1[\text{Free ration village}]_v + \delta_m + \varepsilon_{it} \quad (\text{E.3})$$

where Y_{it} is water orders for household i in month-of-sample t , $1[\text{Exchangeable entitlement village}]_v$ and $1[\text{Free ration village}]_v$ are indicators equal to one if household i resides in an exchangeable entitlement or free ration village, δ_m are month fixed effects, ε_{it} is an error term, clustered at the village level, and the sample consists only of control households.

Table E.2 presents the results. There is no difference between control water orders across the three arms. Point estimates are very small (0.4 or 0.5 litres per month), and not statistically different from zero. Since the types of offers and number of customers ordering water across the three arms are very different, this result implies that neither spillovers between treatment and control households nor seller capacity constraints are likely to explain lower control-group orders in treatment villages.³⁰

³⁰Beyond this evidence from our data, we have no independent reason to suspect capacity constraints — delivery vans had lots of room, treatment plants could easily serve demand, and no concerns were raised by our implementation partner.

Table E.2: Effect of village treatment type on control household water orders

	Orders in litres
Exchangeable entitlement controls	0.51 (1.02)
Free ration controls	0.39 (0.92)
N	218,003
Dependent Variable Mean	2.818

Notes: This table presents a test of differences between water orders among control households only between our three treatment arms, estimated using Equation (E.3). The price arm is the omitted category. The regression includes month fixed effects, and standard errors are clustered by village.

Finally, as we discuss in Section 5, households are not re-selling clean water, so this cannot explain reductions in orders among the control group. We also see no evidence of a consumption kink between consumers paying full price, vs a 10% discount, as seen from the bottom-right panel of Figure 3. If anything, full price consumers order slightly more (conditional on buying anything) than do those with a 10% discount.

More broadly, our experience in the field suggests the marketing effort explanation is more likely. In treatment villages, our implementation partner was guaranteed more consumers because of the discount, free ration, or entitlement cards that were provided to a meaningful number of households. The research team fully reimbursed the implementing partner for all subsidies, raising revenues in these villages. As a result, it is highly likely that the implementing partner expended less marketing and sales effort among control customers in treatment villages relative to pure control villages. For all these reasons, although we cannot rule out either of these explanations, our hypothesis is that seller effort may be more important than spillovers or behavioral responses by control consumers.

Nevertheless, it is straightforward to de-bias our WTP measure by re-estimating the demand curve with the assumption that under equal effort / no spillovers (i) the number of consumers at the highest price would rise to match levels in the pure control, and (ii) consumers at all price levels would use additional water equal to the estimate from Column (1) of Table E.1. Doing so leads to an adjusted WTP of INR 153, up from INR 132 in the main experiment sample.

F Calculating profitability

To calculate the net present value of the profits from selling clean water, we use data from Spring Health on costs, as well as a series of assumptions, enumerated in Table F.1. We begin by computing annual revenues and variable costs for clean water take-up levels ranging from 10 to 100% of the households in a representative village. We use the extensive-margin demand curve to identify the price associated with each take-up share. The number of consumers at each take-up level is simply the number of households in our representative village multiplied by the take-up share. Annual revenue is thus simply $\text{Revenue} = \text{Price} \times \text{Consumers} \times \text{Per-consumer usage} \times 12$, and, annual total variable costs are: $\text{Variable cost} = \text{Per-consumer cost} \times \text{Consumers} \times 12$.³¹ Finally, we compute the net present value of costs as the up-front cost of installing a water treatment plant plus annual fixed costs and annual total variable costs over the assumed life of the plant, and compute the net present value of revenue as annual total revenue over the life of the plant, both discounted using our assumed discount rate. Profits are thus revenues net of costs.

³¹Our cost data are monthly, so we multiply by 12 to compute annual costs.

Table F.1: Assumptions for profit calculation

Panel A: Up-front costs	
Item	Cost per plant (INR)
Water purifier	260,700
Tank, motor, & fittings	39,570
Plumber	3,500
Painting	15,000
Plant structure	45,000
Iron plate frame	10,000
Transportation of machine and bottles	3,000
Launching costs	12,000
<i>Total</i>	<i>388,770</i>
Panel B: Monthly fixed costs	
Item	Cost per month (INR)
Employee salaries	5,543
<i>Total</i>	<i>5,543</i>
Panel C: Variable costs	
Item	Cost per customer-month (INR)
Water bottle	21.83
Water bottle stickers	2.92
Entrepreneur commission (incl. water costs)	66.45
Delivery costs	75.31
<i>Total</i>	<i>166.51</i>
Panel D: Assumptions	
Item	Value
Nr. of households	450
Litres per month	237
Discount rate	5%
Plant life (years)	10
Mean villages served by each plant	1.5
Life of water bottle (years)	1
Monthly bottle rental price (INR)	66.67

Notes: This table reports the cost data from Spring Health Pvt Ltd. (2023) and assumptions used in our profitability calculation.

G Calculating cost-effectiveness

To calculate cost-effectiveness metrics, we follow the calculation approach laid out in Kremer et al. (2023) for several clean water provision scenarios. For each scenario, we only consider benefits from reduced mortality among children under 5. We do not measure this treatment effect directly, but instead rely on the Kremer et al. (2023) meta-analysis estimate. For all scenarios, we evaluate the intervention for the rural Odisha context, using data on the per-capita GDP in Odisha, the number of children per household, and the under-5 mortality rate. All cost estimates assume 5 years of water provision. Our scenarios are as follows.

We begin with the Spring Health (full subsidy) scenario. In this scenario, we assume that *all* households in a village receive offers of free Spring Health water. To simulate a setting in which Spring Health or an equivalent has not yet entered, our cost measure includes fixed costs, and our average total costs are socialized across all households who take up clean water. We assume the control group bought zero clean water, thus our intervention take-up rate is the intent-to-treat effect of the free ration offer on take-up plus the average take-up in the control group. We scale this take-up rate by the share of village-months where Spring Health water deliveries were available during our sample period to account for service disruptions.

Second, in the Spring Health (targeted) scenario, we only provide subsidized water to households with children. This lowers the cost of provision per child, but because fewer households in the village will participate in clean water, our average total costs are socialized over a smaller number of households.

Third, our Spring Health (targeted, no FC) scenario assumes that a Spring Health water treatment plant already exists in the village, and we provide subsidies to households with children only. Thus, this scenario removes fixed costs, and uses the intent-to-treat impact on take-up from the free ration arm as our intervention take-up rate.

Fourth, our Spring Health (full service) scenario repeats the (targeted, no FC) scenario, but assumes that there are no water service disruptions (i.e., that Spring Health water is always available).

Finally, we compare these against two chlorine scenarios: point-of-collection chlorine dispensers, as described in Kremer et al. (2011b), and coupons for point-of-use chlorine treatment distributed through maternal and child health services, as described in Dupas et al. (2016). We use data on these two approaches from Kremer et al. (2023), adjusting the relevant parameters for the Odisha context. Because Kremer et al. (2023) does not provide information about disruptions, we conservatively assume this water is always available.

Under each scenario, we compute expected deaths averted per child under-5 as:

$$\text{Deaths averted per child} = \frac{\text{U5 mortality rate}}{100} \times [1 - \text{risk ratio}] \quad (\text{G.1})$$

$$\times \frac{\text{Intervention take-up}}{\text{Meta-analysis compliance}} \times \text{Water availability}, \quad (\text{G.2})$$

cost per DALY saved as:

$$\text{Cost per DALY saved} = \frac{\text{Cost of provision}}{\text{Deaths averted}} \div (\text{DALYs per life}), \quad (\text{G.3})$$

and net benefits per child as:

$$\text{Net benefits} = (\text{DALYs saved} \times \text{per-capita GDP}) - \text{Provision cost}, \quad (\text{G.4})$$

Table G.1 provides the values for each object in the above equations. We summarize the cost per DALY and net benefits in Figure 7.

Table G.1: Details of cost-effectiveness calculation

	Spring Health (full subsidy)	Spring Health (targeted)	Spring Health (targeted, no FC)	Spring Health (full service)	Chlorine (dispensers)	Chlorine (MCH coupons)	Source
DALYs per life	79.25	79.25	79.25	79.25	79.25	79.25	World Health Organization (2020)
Under-5 mortality rate (p.p.)	4.27	4.27	4.27	4.27	4.27	4.27	Ministry of Health and Family Welfare (2022)
Odisha per-capita GDP	1,797	1,797	1,797	1,797	1,797	1,797	Government of India (2023)
Meta-analysis risk ratio	0.77	0.77	0.77	0.77	0.77	0.77	Kremer et al. (2023)
Meta-analysis compliance rate	0.53	0.53	0.53	0.53	0.53	0.53	Kremer et al. (2023)
Intervention take-up rate	0.90	0.90	0.89	0.89	0.36	0.26	SH: Free ration arm; Chl: Kremer et al. (2023)
Water availability	0.88	0.88	0.88	1.00	1.00	1.00	SH: Disruptions; Chl: N/A
Treated children per household	0.4	1.0	1.0	1.0	1.0	1.0	Government of India (2011)
Cost of provision, per child (USD)	261	122	93	93	56	13	SH: Appx. F; Chl: Kremer et al. (2023)
Deaths averted, per child	0.0146	0.0146	0.0145	0.0165	0.0067	0.0048	Eq. (G.2)
DALYs saved, per child	1.16	1.16	1.15	1.31	0.53	0.38	Deaths averted $\times$ DALYs per life
Cost per death averted (USD)	17,878	8,354	6,421	5,631	8,410	2,636	Provision cost / deaths averted
Cost per DALY (USD)	226	105	81	71	106	33	Eq. (G.3)
Net benefits (USD)	1,821	1,961	1,967	2,256	894	673	Eq. (G.4)

Notes: This table reports our cost-effectiveness calculations, including the assumptions required to carry them out. Spring Health (full subsidy) assumes subsidies are given to all households, and fixed costs, which are socialized across all households who consume, are included. In this group, we assume control take-up is zero. Spring Health (targeted) assumes subsidies are only given to households with children, and fixed costs, which are socialized across only households with children who consume, are included. In this group, we assume control take-up is zero. Spring Health (targeted, no FC) assumes subsidies are only given to households with children, and fixed costs are excluded (simulating a situation in which a private firm enters on its own). Spring Health (full service) assumes water is available 100% of the time. Chlorine (dispenser) and chlorine (MCH coupons) information is taken from Kremer et al. (2023), but adjusted to use Odisha's GDP per capita, mortality rate, and number of children per household. Throughout, we follow Kremer et al. (2023) in using 5 years of costs. See the text for equations.

H Deviations from our pre-analysis plan

This experiment was pre-registered with the AEA as AEARCTR-0010545.³² Though we endeavour to follow the PAP as closely as possible, we enumerate our deviations below:

- **Analysis.** PAP equations (1), (2), and (3) use purely cross-sectional variation (plus LASSO-selected controls). In the paper, we present results only using Equation (4), a panel specification, for the sake of parsimony. Results using the cross-section are quantitatively similar.
- **Analysis.** In the PAP, we pre-specified using pure control villages as robustness checks in analysis. Because our analysis includes village fixed effects, our treatment effects are not identified off of between-village differences, so adding pure control villages to the regression would not change our estimates. We therefore only analyze these villages in Appendix E.
- **Analysis.** On page 9 of the PAP, we pre-specified using post-double-selection LASSO to choose controls. We do not include controls (other than fixed effects) for two reasons. First, since we submitted our PAP, new work (Cilliers, Elashmawy and McKenzie, 2024) has arisen, arguing that selection of controls via LASSO in RCTs is essentially useless. Moreover, even if we wanted to add controls, we only have control variables for the subset of households that participated in our survey, so we cannot add household observables into the administrative data regressions for the vast majority of households.
- **Analysis.** PAP section 4.2 proposes estimating the price elasticity of demand, both separately for price and exchangeable entitlement arms (Equation (6)) and jointly, with an interaction term for being in the exchangeable arm (Equation (7)). Given the large number of zeroes in our consumption data, per Chen and Roth (2024), this log-log specification does not deliver the quantity of interest. Following this paper's guidance, we therefore do not present these specifications and instead focus on the extensive-margin effects and levels effects documented in Table 1.
- **Analysis.** PAP Equation (8) proposes estimating the effects of exchangeable entitlements vs. discounts in a point-by-point manner. This is effectively subsumed by Table 1, so we omit it here.
- **Analysis.** The PAP proposes estimating the effect of our offers on water shortages. In our survey data, households report they ran out of water (across all uses) in the prior week only 2% of the time. As a result, there is no margin for adjustment on this variable, so we omit it from our analysis.
- **Analysis.** PAP equations (10) and (11) propose instrumenting for the quantity of water ordered with water offers. In Table 3 (and its variants), we instead use whether the household reports drinking any clean water for two reasons. First, this ensures that the endogenous variable that comes from the same survey as the outcome variable.

³²The registry entry is available from <https://www.socialscisceregistry.org/trials/10545>.

Second, the effect of drinking any clean water is easier to understand than the effect of ordering one unit of clean water, which must be rescaled to be meaningful. We therefore prefer this endogenous variable.

- **Analysis.** In the PAP, Equations (10) and (11) use all sub-treatment arms in the first stage. We instead present pooled IV estimates for the free ration, exchangeable entitlement, and price arms separately, to measure differential local average treatment effects of each offer type.
- **Analysis.** PAP section 4.3.3 proposes “medium-run” health effects, which use only the health data from endline. As we show in Figure 4, water demand is stable throughout the study. Furthermore, our endline data are collected well after the end of water distribution for the first two implementation waves in our sample, and, as we show in Appendix Table D.5, health treatment effects disappear in the absence of water distribution. Thus, this endline-only exercise is unlikely to add meaningful information and we omit it here.
- **Analysis.** PAP section 4.6 proposes a series of heterogeneity analyses. First, PAP Equation (15) proposes heterogeneity on water quantity by household size, number of children, household income, and quality of drinking water options, all measured at baseline; above- vs. below-median liquidity constraints; an indicator for whether agriculture is the main source of income; whether a household ever wanted to take a loan but was unable to; above- vs. below-median consumption; and above- vs. below-median assets. This equation also proposes a cross-sectional regression specification. We limit the set of covariates to four: an indicator for having any children below 5 in the home; above- vs. below-median household head education; above- vs. below-median income; and an indicator for ever reporting using piped water for drinking during the experiment. We do this for several reasons. First, we did not conduct a baseline survey prior to treatment implementation, and many of the variables we intended to use in heterogeneity could plausibly be affected by treatment. Second, given the high take-up in both the free ration and exchangeable entitlement arm, there is limited scope for heterogeneity. In the interest of parsimony, we therefore present heterogeneity only along a few key covariates.

Furthermore, rather than using PAP Equation (15), which proposes estimating heterogeneous treatment effects on each sub-treatment, we instead measure how WTP for clean water varies with our four covariates. Because take-up is close to 100% for the exchangeable entitlement and free ration arms, there is no meaningful heterogeneity in covariates, so we focus on the price arm. Estimating heterogeneity in WTP rather than demand at each price point enables us to more parsimoniously summarize the impact of household characteristics on demand.

We present the results in Figure 5.

As discussed above, we omit PAP Equation (16) due to concerns about log-log specifications with zeroes.

PAP Equation (17) – and the subsequent un-numbered equation – propose heterogeneity on intent-to-treat effects and local average treatment effects of clean water on

health. Here, we pre-specified heterogeneity by the number of children in the household, household income, quality of drinking water measured at baseline, and whether the household treats their drinking water at baseline. We replace these covariates with the same set that we use in our demand heterogeneity analysis, namely an indicator for having any children below 5 in the home; above- vs. below-median household head education; above- vs. below-median income; and an indicator for ever reporting using piped water for drinking during the experiment. We present the results in Appendix Tables D.4. Because we find limited evidence of heterogeneous ITT effects on health, we omit heterogeneous IV effects on health for the sake of parsimony.

- **Multiple hypothesis testing.** In PAP Section 4.7, we outline multiple hypothesis testing corrections for our health outcomes. We did not measure outcomes for children, so we omit this. We present FDR adjusted p -values for whether any household member was sick in the past week (which is essentially a symptom index), health expenditures, and whether anybody in the household missed work due to illness in Table D.3.
- **Information intervention.** In PAP Section 5, we propose testing the impacts of an information intervention on household water demand. We will conduct this analysis separately, and thus do not include it here.